Q-Learning-Based Multi-Objective Path Planning for AAV Parcel Delivery Through Public Transportation Vehicles

Mohammed Rahmani, Sidi Mohammed Senouci, *Senior Member, IEEE*, Florian Delavernhe, and Marion Berbineau, *Member, IEEE*

***Abstract*—Combining Autonomous Aerial Vehicles (AAVs) with public transportation systems offers a novel approach to enhancing urban last-mile delivery efficiency. However, current research lacks comprehensive solutions for optimizing AAV path planning that can adapt to real-time changes in public transportation schedules while avoiding dynamically changing high-interference zones, ensuring consistent drone connectivity, and accounting for limited drone battery life. This paper addresses these gaps by optimizing and balancing multiple objectives, such as ensuring early arrival, minimizing energy consumption, and reducing signal interference. To achieve these goals, we propose a Q-learning-based algorithm for real-time path optimization, ensuring path adaptability to random changes in interference zones and public transportation schedules, and compare its performance against multi-objective A* algorithm variants. Results show that Q-learning achieves superior early arrival times, energy efficiency, and reduced interference, with a nearly 100% success rate in package deliveries, compared to 60-80% for A*. These findings demonstrate the robustness and reliability of Q-learning in AAV-Aided Public Transportation systems, making it a valuable solution for real-world urban delivery challenges.**

***Index Terms*—AAV delivery, public transportation, early arrival path, energy consumption, interference, Q-learning.**

I. Introduction

THE growing integration of autonomous aerial vehicles (AAVs) into civilian applications such as surveillance [1], wireless communications [2], [3], and package delivery [4] has opened numerous new research directions. Among these, AAV-based logistics has emerged as a transformative solution with the potential to revolutionize last-mile delivery systems, significantly reducing operational costs, minimizing delivery times, and mitigating environmental impact through optimized energy usage.

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Mohammed Rahmani is with the Université Bourgogne Europe, DRIVE UR 1859, 58000 Nevers, France, and also with COSYS, Université Gustave Eiffel, 59650 Villeneuve-d'Ascq, France (e-mail: Mohammed_Rahmani@u-bourgogne.fr).

Sidi Mohammed Senouci and Florian Delavernhe are with the Université Bourgogne Europe, DRIVE UR 1859, 58000 Nevers, France (e-mail: sidi-mohammed.senouci@u-bourgogne.fr; Florian.Delavernhe@u-bourgogne.fr).

Marion Berbineau is with COSYS, Université Gustave Eiffel, 59650 Villeneuve-d'Ascq, France (e-mail: marion.berbineau@univ-eiffel.fr).

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Recently, both logistics companies like Amazon (see [5]) and researchers have shown increased interest in using AAVs to develop efficient last-mile delivery systems in terms of timely delivery, energy efficiency, etc.

Approaches for AAV-based delivery have included direct deliveries from warehouses to customers [6], [7], which are fast but limited by AAVs' battery life, and truck-assisted AAV deliveries [8], [9], which extend range but incur high costs for maintenance and fuel. An innovative alternative to mitigate these expenses while broadening AAVs' delivery scope involves utilizing public transportation vehicles during their regular routes, with AAVs landing on their roofs for intermediate transport [10], [11]. This AAV-aided public transportation system (AAV-APT) not only reduces costs but also aligns with sustainable urban mobility goals, leveraging existing infrastructure to minimize carbon emissions and traffic congestion. By integrating AAVs with public transit networks, logistics companies can achieve scalable, environmentally friendly delivery solutions that enhance urban logistics systems.

Numerous studies have extensively explored the multi-faceted challenges associated with AAV-APT system, highlighting path planning as a central and intricate issue. The core challenges in this context are multifaceted. Suppliers often prioritize fast delivery to meet customer demands, requiring the planning of early arrival paths instead of the shortest routes, which can lead to trade-offs between speed and efficiency. They also strive to serve customers located far from warehouses, expanding the delivery area and increasing energy consumption while requiring reliable AAV connectivity. Public transportation vehicles, operating on fixed routes and schedules beyond logistical control, further complicate delivery coordination. Additionally, unpredictable factors such as traffic congestion and mechanical issues introduce variability in vehicle travel times. AAV route optimization must balance the need for rapid delivery through direct flights and energy efficiency by leveraging public transportation. These complexities are compounded by the necessity of maintaining uninterrupted cellular connectivity for AAVs, which is often challenged by interference from terrestrial user devices and environmental factors.

Offline path planning, such as planning before the AAV departs from the warehouse, is often insufficient due to the stochastic and time-dependent nature of public transportation networks and interference zones. As a result, online path planning has emerged as a superior alternative. It allows AAVs to dynamically adjust their trajectories based on real-time data, improving adaptability and efficiency.

Hence, the central concept of this paper revolves around empowering the AAV to autonomously adapt its flight path in real-time. This adaptability accommodates fluctuations in public transportation schedules and variations in connectivity quality, all while considering the AAV's limited battery capacity. This process is referred to as online path planning with multi-objective optimization. To the best of our knowledge, there is limited research that explores this process within a stochastic, time-dependent public transportation network, ensuring prompt delivery, energy efficiency, and sustained AAV connection quality.Our work addresses this gap by employing Q-Learning [12], a robust method capable of optimizing multiple objectives in unmodeled and dynamic environments, such as dynamic public transportation networks and fluctuating connectivity quality, both spatially and temporally. Unlike traditional path planning algorithms, our approach dynamically adapts to environmental fluctuations and optimizes multiple objectives simultaneously, including minimizing early arrival times, energy consumption, and interference. The effectiveness and robustness of the proposed Q-Learning approach are quantitatively demonstrated through key experimental metrics, including path optimization efficiency, energy consumption, delivery timeliness, and interference mitigation. These metrics are used to compare the performance of Q-Learning against traditional offline path planning algorithms, showing its adaptability and superior performance in real-time, dynamic conditions. By utilizing real-time feedback from the system, our algorithm can effectively address the unpredictable nature of public transport schedules and fluctuating connectivity, determining the optimal AAV path and integrating AAV flight with public transportation to enhance overall delivery efficiency. The primary contributions of this paper can be outlined as follows:

- We introduce a multi-objective path planning problem for AAVs in the AAV-APT system, aiming to minimize early arrival times, energy consumption, and accumulated interference. While existing works [11], [13], [14] focus on basic AAV-APT architectures and single or dual-objective path planning, our contribution lies in adopting a multi-objective optimization approach. By framing the problem as an online challenge, we enable real-time decision-making, allowing the system to adapt dynamically to fluctuating public transportation schedules and high-interference zones.
- Building on this novel problem formulation, we propose a multi-layer model that integrates key components, including the public transportation network, AAV flight range, energy consumption, ground user mobility patterns, and an interference coverage map. This integrated approach supports accurate path planning and enables more precise optimization of AAV trajectories compared to traditional methods, which often overlook these complexities.
- To address this complex problem, we leverage reinforcement learning (RL), specifically Q-learning, marking its first known application to multi-objective path planning in AAV-APT systems. We introduce an innovative reward function and customize the state representation to incorporate AAV-specific attributes, public transportation schedules, user mobility patterns, and interference levels. This comprehensive adaptation ensures that Q-learning can effectively handle the dynamic, real-time environment of the AAV-APT system.
- Finally, we validate our approach through extensive simulations, comparing the proposed Q-learning method with traditional offline path planning algorithms, including A* and its modified path-repair variations. Results consistently demonstrate that our Q-learning approach outperforms these methods, ensuring successful deliveries even under disrupted public transportation schedules and high-interference conditions. This highlights the robustness and superior performance of our method in dynamic, real-time environments.

The remainder of this paper is organized as follows. In Section II, we discuss the related works. The system model and problem formulation are described in Section III and IV. Section V presents the proposed Q-learning algorithm for finding the multi-objectives optimal path. In Section VI, simulation results are analyzed and conclusions of the paper are drawn in VII.

II. Related Work

In recent years, there has been a significant increase in interest, in both academic literature [15] and industry reports [5], concerning the utilization of autonomous aerial vehicles (AAVs) for last-mile deliveries. Although the idea of using AAVs for transporting parcels is not new, recent technological advancements have transformed it into a practical solution for addressing the growing demand for fast and efficient deliveries.

Considerable research has been conducted to develop sustainable AAV-based last-mile delivery through various designs and approaches. These include AAV-only delivery models, where autonomous AAVs transport goods directly from warehouses to customers [6], [16], and AAV-truck hybrid models, where trucks serve as mobile warehouses that transport AAVs and packages to designated points, from which AAVs complete the final delivery leg [8], [9]. Recent studies have also explored AAV swarms for urban logistics, focusing on aspects such as task assignment, route planning, and energy efficiency. For instance, [17] proposes dynamic multi-dimensional assignment and routing methods to coordinate swarms under various delivery scenarios. Similarly, [18] investigates how swarm intelligence, combined with smart city infrastructure, can enhance drone logistics while addressing challenges like communication, battery limitations, and urban safety. In another study, [19] presents a hybrid genetic algorithm combined with an enhanced Dijkstra to minimize energy consumption and ensure reliable communication during multi-AAV delivery operations. Furthermore, a hybrid solutions that leverage the mobility of public transport vehicles to transport AAVs on their roofs for package delivery [11]. Such approaches have inspired numerous research directions, particularly on AAV path planning, which aims to optimize delivery objectives such as energy efficiency, cost, and connectivity while adapting paths to existing infrastructure.

Research in AAV path planning addresses challenges like time efficiency, obstacle avoidance, energy consumption, cost-effectiveness, and connectivity. For instance, the study in [20] introduced a strategy to optimize the time efficiency of path planning for multiple autonomous AAVs tasked with collecting data from distributed Internet of Things (IoT) devices, employing multi-agent reinforcement learning (MARL). Reference [21] conducted a comprehensive comparative analysis of algorithms for AAV path planning, focusing on obstacle avoidance to ensure safe AAV operations across various civilian domains.

Given the significant impact of energy constraints on AAVs, considerable attention has been directed towards enhancing energy efficiency. Several papers have explored energy-efficient path planning strategies [22], [23], [24]. In reference [22], the authors used optimal control theory to devise a path that efficiently covers and scans all forest trees while conserving energy. Similarly, another addressed AAV path planning as a traveling salesman optimization problem, aiming to minimize energy consumption using a genetic algorithm [23]. Furthermore, an analytical survey of various path planning methodologies, specifically examining their impact on AAV energy efficiency, has been provided [24]. Additionally, researchers tackled the AAV-aided last-mile delivery problem with shared depot resources, integrating tactical decisions about fulfillment centers and fleet size planning into a unified framework while considering non-linear energy consumption for AAV batteries [25]. Another study aimed to minimize total waiting times by optimizing AAV paths from warehouses to customers, accounting for non-linear energy consumption and uncertain AAV flight times through robust optimization approaches [26].

Connectivity-aware path planning is equally crucial, especially for cellular-connected AAVs. One introduced an interference-aware path planning approach for multiple cellular-connected AAVs flying at a fixed altitude, aiming to balance path length, latency, and interference with the ground network [27]. Another addressed the optimal path planning problem for a AAV navigating from a source to a destination in a 2D space, aiming to maximize communication quality within existing cellular coverage [28]. Furthermore, researchers aimed to minimize interference and enhance connectivity for a cellular-connected AAV during its journey while ensuring robust communication with ground base stations (GBSs) [29]. In a recent study on optimizing navigation for AAVs in urban environments [30], the significance of connectivity quality and energy consumption is emphasized. The study proposes a multi-criteria optimization approach that balances minimizing energy consumption through trajectory length reduction while maximizing connectivity quality along the AAV's path, demonstrating promising results with the proposed ϵ-constraint method and modified RRT*FN algorithm.

While many studies focus on AAV operations in isolation, there is growing interest in AAV collaboration with other infrastructures. For instance, one method utilizes K-means clustering and the traveling salesman problem to develop paths that extend delivery range while optimizing truck routes [31]. Moreover, a label-setting algorithm for reliable AAV path planning within a parcel delivery system involving AAVs and public transportation vehicles has been proposed in [13].

In addition to technological advancements and algorithmic solutions, considering the practical implications of implementing AAV path planning in urban settings is vital. Regulatory compliance, safety concerns, and integration with existing infrastructure are critical factors. As AAVs evolve, global regulators strive to keep pace with their capabilities; understanding international drone legislation is essential for identifying countries that are open to delivery drones and setting precedents for others [32]. The rise of e-commerce and international trade has heightened the importance of logistics, including drone technology, which enhances delivery efficiency [33]. However, privacy and safety concerns must be addressed, as the implementation of drone delivery often outpaces existing legislation [34]. Aligning proposed methods with regulatory frameworks and integrating AAVs into urban logistics networks is essential for the success of AAV-based delivery systems.

Based on a comprehensive literature review and to the best of our knowledge, no previous studies have specifically addressed multi-objective path planning for AAV parcel delivery through a AAV-APT system. Such a path should dynamically adapt to real-time changes in transport networks, adhere to logistical requirements such as early arrival delivery processes, prioritize energy efficiency, and maintain robust connectivity for AAV operations. The current study aims to fill these research gaps by introducing an online path planning approach that is time-adaptive, prioritizes early arrival, ensures energy efficiency, and is connectivity-aware for AAVs engaged in AAV-APT parcel delivery missions. To clearly highlight the research gaps and demonstrate how our work addresses these challenges, we present a detailed literature survey analysis in Table I. This table summarizes prior work, identifies their limitations, and outlines the specific contributions of our study in bridging these gaps.

III. System Model

In this work, the online path planning problem seeks to leverage both public transportation and AAV capabilities to optimize the delivery of parcels to end-users. This multifaceted problem entails calculating optimal paths for AAVs while focusing on several objectives. Foremost among these is the maximization of the Signal-to-Interference-plus-Noise Ratio (SINR) for cellular-connected AAVs during their missions, necessitating the minimization of interference signals originating from ground User Equipment (UEs) and other Base Stations (GBSs). This optimization is pivotal to ensure robust and high-quality communication throughout the AAV's flight. Additionally, we strive for energy-efficient AAV operation, as this objective also bolsters the overall sustainability and efficacy of the parcel delivery system. Efficiency is further augmented by guaranteeing quick arrivals at customer destinations, a pivotal factor in customer satisfaction for parcel deliveries. Furthermore, the paths obtained must adapt to the

TABLE I

LITERATURE SURVEY, RESEARCH GAPS, AND OUR CONTRIBUTIONS

Theme	Refs.	Prior Work Focus	Research Gaps	How This Work Addresses the Gaps
AAV Delivery Architectures	[6, 16, 8, 9]	AAV-only and truck-AAV hybrid delivery systems	Do not handle real-time public transport schedule variations or urban dynamics	Integrates AAVs with public transport for real-time adaptive delivery responding to schedule fluctuations
Dynamic Task Assignment in Multi-AAV Systems	[17, 18, 19]	Swarm coordination and dynamic task assignment	Mostly static assumptions; lack real-time adaptive task allocation under urban interference	Introduces Q-learning-based dynamic task assignment adaptable to stochastic urban conditions; groundwork for multi-AAV coordination
Path Planning Under Operational Constraints	[22, 23, 24, 25]	Energy-efficient routing and cost minimization	Focus on static, single-objective models; miss multi-objective real-world variability	Develops multi-objective Q-learning optimizing energy, delivery time, and interference in real-time
Connectivity-Aware AAV Navigation	[27, 28, 29, 30]	Communication quality and interference management	Few integrate connectivity optimization with dynamic interference and coverage changes	Proposes connectivity-aware path planning balancing interference, latency, and AAV constraints dynamically
Urban Integration & Regulatory Compliance	[32, 34, 33, 13, 11]	Regulation, safety, and infrastructure integration	Lack of dynamic incorporation of regulatory and safety constraints in path planning	Embeds real-time adaptation to transport delays and regulations for safe AAV urban operations

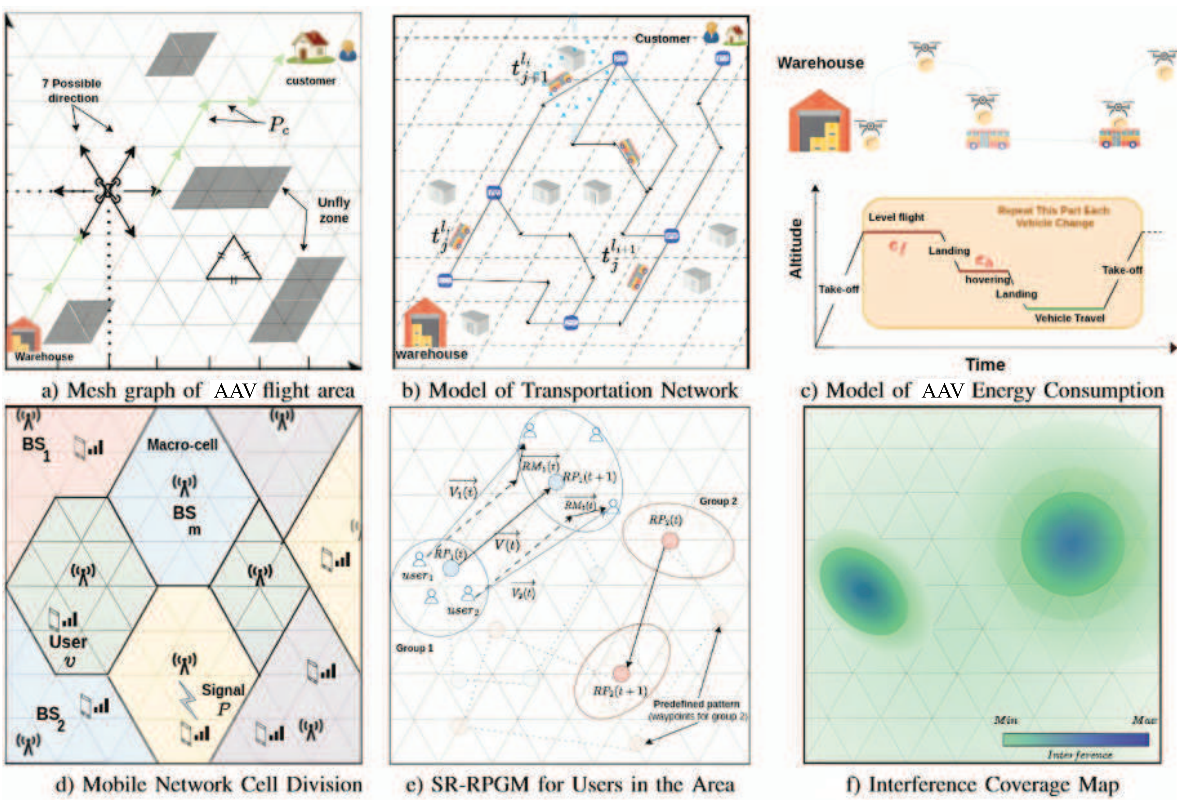


Fig. 1. Multi-layer model for AAV-Aided public transportation in urban areas delivery freight system.

dynamic nature of public transportation schedules, enabling seamless adjustments to schedule fluctuations and ensuring the integration of AAVs with existing transportation networks for maximum efficiency.

To provide a comprehensive representation of critical parameters essential for AAV path planning and analysis, this section introduces a multi-layer model designed to accommodate the dual mode of travel employed by the AAV, encompassing both flight and bus transportation. As depicted in Figure 1, this model covers the mesh graph of AAV flight area, the model of transportation network, the model of AAV energy consumption, mobile network cell division, semi-random Reference Point Mobile Group model (SR-RPGM) for users in the area, and an interference coverage map.

A. AAV Flight Area

Comprehending the AAV flight area is of paramount significance in effective path planning, particularly within urban settings where parcel deliveries are a critical concern. This comprehension involves a multifaceted approach, encompassing tasks such as identifying no-fly zones, meticulously delineating permissible flying areas, and intricately plotting feasible flight directions to ensure the utmost efficiency and safety of operations. As vividly depicted in Figure 1(a), our approach employs a meticulously crafted mesh graph $G_1(V_1, E_1)$ to achieve this objective. Each node $v \in V_1$ in this graph is associated with a physical position denoted as (x_v, y_v). The set E_1 represents the links between pairs of nodes, symbolizing connections through direct AAV flights:

$$E_1 = \langle v, v', l_0, t_0\rangle, |v, v' \in V_1$$

Here, the label l_0, t_0 is used to represent AAV flights.

In essence, we partition the flight area into distinct segments, maintaining a uniform distance between each position node strategically placed at the center of every rhombus. These nodes are seamlessly connected by edges, facilitating smooth navigation between adjacent rhombus segments. In our study, we specifically employ a 6-degree mesh graph, denoting a AAV flight network. This signifies that each node has six neighboring nodes at the same uniform distance, affording AAVs six potential directions of movement, including the option to remain stationary when necessary. Additionally, our model takes into consideration no-fly zones, where obstacles or restricted fly zones may exist.

B. Model of Transportation Network

As our system also includes the public transportation network, in this subsection, we model the transportation network using a graph denoted as $G_2 = (V_2, E_2)$, see Figure 1(b). Here, V_2 is a set of nodes representing the vehicle stops, and E_2 is a set of links connecting pairs of vehicle stops:

$$E_2 = \{\langle v, v', l_i, t_j\rangle, |, v, v' \in V_2, V_2 \in V_1, l_i \in L, t_j \in T^{l_i}\}$$

In this context, l_i denotes the public transport line number i, and L indicating the set of all these lines. Furthermore, t_j represents trip number j on one of the lines, with T^{l_i} being the set of trip journeys for line l_i. We also denote the traversal time between two nodes as $\theta^{l_i t_j}_{vv'}$. In addition, for the sake of simplicity, we assume that each bus station $v \in V_2$ is located at one of the nodes $v \in V_1$ in the graph representing the AAV flight area (G_1). Additionally, each edge in graph G_2 is a concatenation of successive edges in G_1, where each $e \in E_2$ is defined as follows:

$$e = \{e_i \oplus e_{i+1} \oplus \ldots \oplus e_{i+k}, |e_i \in E_1\}$$

Within the realm of traversal time, we introduce two functions that play a pivotal role in our model. These functions describe the arrival and departure times of a trip t_j at a given stop v as $\alpha(t_j, v)$ and $\tau(t_j, v)$, respectively.

In this case, $G(V, E)$ be an undirected graph that combines $G_1(V_1, E_1)$ and $G_2(V_2, E_2)$. Specifically, $V = V_1$ and $E = E_1 \cup E_2$.

Definition 1: We define path p_c from the warehouse $w \in V$ to the consumer $c \in V$ in G as a sequence of links $(v_z, v_{z+1}, l_i, t_j) \in E$ as:

$$p_c = \left\{(w, v_1, l_i, t_j), (v_1, v_2, l_{i'}, t_{j'}) \ldots (v_{m-1}, c, l_m, t_k)\right\}$$

where each consecutive couple of links in this path should satisfies:

$$\forall (v_{z-1}, v_z, l_i, t_j), (v_z, v_{z+1}, l_{i'}, t_{j'}) \in P_C :$$
$$\alpha(l_i, t_j, v_z) \leq \tau(l_{i'}, t_{j'}, v_z)$$

In the rest of the paper, we define P_c as set of all possible paths from w to c.

Definition 2: On one hand, to simplify the modeling process during AAV flight, we assume a average speed between two nodes, within the range of $0 < V \leq V_{\max}$, where $V_{\max}$ represents the AAV's maximum speed. As a result, the time required for the AAV to travel between any two nodes is computed based on this average speed, instead of accounting for speed fluctuations during the flight. On the other hand, when the AAV operates in conjunction with public transportation, its travel time is assumed to be equivalent to the time taken by the bus between two positions. The AAV's time-varying coordinates are denoted as: The AAV's time-varying coordinates are denoted as:

$$p_t = (x_t, y_t), t \in [0, T_c[$$

where T_c denotes the mission completion time, which refers to the arrival of the drone to customer c.

C. AAV's Energy Consumption Model

To estimate power consumption within AAV-APT delivery systems, this subsection introduces additional modeling concepts. As shown in Figure 1(c), we divide the AAV's energy usage into two primary phases: hovering and flight. During periods when the AAV travels on top of a public vehicle, we assume zero energy consumption. It's important to note that we do not consider the power consumption of the controller in this analysis, as well as the takeoff and landing power consumption, as their consumption is negligible compared to other stages. Let e_f and e_h denote the energy consumed by the AAV during flight and hovering, respectively. To establish a connection between the AAV's energy consumption and the duration of its travels, we introduce e_c. that serves as a representation of energy consumption along a given path p_c, and its computation is detailed as follows:

$$e_c = \sum_{(v,v',l_i,t_j)\in p_c} (\tau(t_j, v) - \alpha(t_{j-1}, v))e_h + \theta^{l_i t_j}_{vv'} e_f \tag{1}$$

D. Communication Model

To ensure uninterrupted communication for the AAV during its flight, it is essential to establish and maintain a connection with one of the available Base Stations (BSs), collectively referred to as the ensemble M of BSs. In our research, we partitioned the service area into distinct non-heterogeneous mobile cells, as depicted in Figure 1(d). Each cell is served by one BS, denoted as $m \in M$. Furthermore, within each cell and at any given time instant t, there exists an ensemble of mobile users, denoted as v, with their spatial distribution represented by Φ.

To model the mobility of users within an area over time, let's consider an example of university students moving around the campus simultaneously. Their movements often occur in groups and follow logical paths between dedicated locations within the campus, such as transitioning from dormitories to classrooms, and then to restaurants, among other destinations. For that, we employ a semi-random Reference Point Mobile

Group model (RPGM), as illustrated in Figure 1(e), to simulate their mobility patterns for multiple groups within the city. This model is particularly well-suited for capturing the distribution and movement behaviors of specific user groups in an urban environment. Additionally, our model accounts for the logical movements of citizens between various points of interest in the city throughout the day.

The mobility of these users leads to variations in their distribution across different areas, resulting in varying levels of connectivity quality in different regions, as depicted in Figure 1(f). In this work, the received Signal-to-Interference-plus-Noise Ratio ($SINR^t$) serves as a measure of the quality of connectivity for the GBS-AAV communication link. It involves evaluating the total received signal power at the AAV originating from a designated GBS, denoted as m, within a specified time slot t. This calculation considers various contributing factors, including the transmitted signal of interest, potential interference from neighboring terrestrial transmitters, and thermal noise. Consequently, the instantaneous Signal-to-Interference-plus-Noise Ratio (SINR) for the AAV is determined by the following expression:

$$SINR^t_{\text{aav}} = \frac{P^t_m G^t_m}{I^t_{\text{aav}} + \delta^2} \tag{2}$$

where P^t_m is the transmit power of GBS m at time t, δ^2 denotes the thermal noise level, and the details of its calculation are given in Section A1 of the Appendix, G^t_m the channel power gain, between transmitter m and the AAV at time slot t represents a critical element in assessing communication quality. This gain is given by:

$$G^t_m = L^t_m F^t_m \tag{3}$$

where (L^t_m) is the large-scale slow fading, encapsulating path-loss and shadowing effects (see Section A2 of the Appendix for more details), and (F^t_m) denotes the small-scale fast fading (see Section A3 of the Appendix for more details), simulating rapid signal fluctuations due to factors like multi-path propagation. Finally, I^t_{aav} represents the interference experienced by the AAV at time t. It can be modeled as the sum of cumulative interference from other users, which depends on the spatial distribution of users, as well as the cumulative of signals received from other base stations. This is expressed by the following equation:

$$I^t_{\text{aav}} = \sum_{u \neq \text{aav}}^{u \in \Phi} P^t_u G^t_u L(\left\| X^t_{\text{aav}} - X^t_u \right\|) + \sum_{i \neq m}^{M} P^t_i G^t_i \tag{4}$$

where $L(\left\| X^t_{\text{aav}} - X^t_u \right\|)$ represents the path loss function that accounts for how the signal strength between the AAV and user u varies with the Euclidean distance between the AAV location (X^t_{aav}) and user's location u (X^t_u) Finally, for completing our model, we introduce the concept of double-weights, denoted as Ω, associated with pairs of nodes. These double-weights encompass two crucial parameters: traversal time ($\theta^{l_i t_t}_{vv'}$) and the average SINR ($\varepsilon^{l_i t_t}_{vv'}$) experienced by the two nodes under consideration.

$$\Omega = \{ \prec \theta^{l_i t_t}_{vv'}, \varepsilon^{l_i t_t}_{vv'} \succ | v, v' \in V, l_i \in L, t_j \in T^{l_i} \}$$

IV. Problem Formulation

In this section, we present the core framework for the multi-objective path planning problem and describe the methodologies used to solve it. First, in subsection IV-A, we define the problem and outline the key objectives we aim to optimize. Next, in subsection IV-B, we introduce the Markov Decision Process (MDP) to model the AAV's decision-making process in the dynamic environment. Finally, in subsection IV-C, we discuss the concept of Q-learning, a reinforcement learning approach, which is applied to solve the problem with multi-objective optimization, enabling the AAV to adapt its path based on real-time conditions.

A. Problem Definition

In the context of optimizing the path for delivering parcels to customers using a AAV-APT system, the objectives include achieving early arrival time, minimizing energy consumption, and maximizing SINR value (accomplished by minimizing the total received interference), all while accommodating random changes in bus schedules and the stochastic and time-dependent nature of the public transportation network, intertwined with the dynamic evolution of terrestrial user device locations. The challenge stems from conflicting objectives, making it impossible to find a single optimal solution for all objectives. For instance, while direct flights offer the quickest travel time, they are also the most energy-consuming (as opposed to the AAV being transported by a bus) and do not consider potential interferences. Thereafter, the goal is to find a set of solutions that represent the best trade-offs among these objectives, rather than seeking a single and unique solution, with the decision-maker selecting the most suitable route according to implicit needs. However in an online decision-making context, where the path of the AAV is dynamically optimized in reaction to stochastic events (e.g., delays in bus timetable), it is not realistic to consider a decision-maker selecting a new solution from a different set after each update. Thereafter, our approach is the following. We employ a weighted-sum approach that optimize the offline problem, offering a set of efficient solutions. After that, the decision-maker agent (which can be a complex algorithm ensuring the scheduling of multi-AAV deliveries) selects a solution which best aligns with their preferences. Thereafter, the online adjustments are made accordingly to the weights used to generate the selected solution, which removes the necessity for an online handmade selection while keeping previously expressed preferences. For a given graph $G(V, E, \Omega)$, t_0 (the instant when the customer places an order), and E_0 (the initial energy), the optimization problem is formulated as follows:

$$p^*_c = \arg_{P_c} \min \left(\gamma\alpha(., T) + \partial e_p + (1 - \gamma - \partial) \sum_{t=t_0}^{T_c} I^t_{aav} \right) \tag{5}$$

Here, P_c represents all possible paths from the warehouse to the customer. Different values of the parameters γ and ∂ are tested offline, with $\gamma + \partial < 1$, but are fixed for online optimization. These parameters determine the trade-off between early arrival time, energy consumption, and minimizing the total received interference.

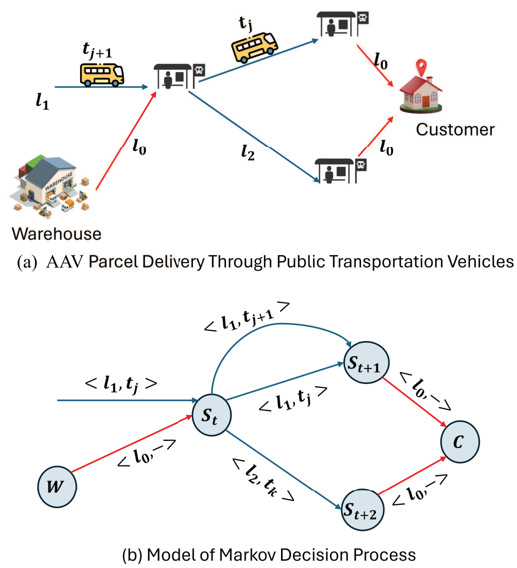


Fig. 2. Markov decision model for AAV parcel delivery through public transportation vehicles.

B. Markov Decision Process (MDP) Framework

To tackle the optimization of this objective, we utilize a Markov Decision Process (MDP) to formulate the problem. The foundation of this AAV path planning system is based on a Markov Decision Process (MDP), characterized by the tuple $\langle S, A, T, R \rangle$. In this context:

- S represents a finite set of states that correspond to nodes within a mesh graph.
- The set A comprises a finite set of actions, each described by a tuple $< l_i, tj \;\; >$, which encompasses public transportation trips, AAV flying trips, and even hovering actions in place.

The transition probability function $T : S \times A \times S \rightarrow [0, 1]$ encapsulates the probability that a AAV, through action $a =< l_i, t_j >$, moves from state s_t to state s_{t+1}. Additionally, the reward function $R : S \times A \rightarrow R$ quantifies the real-time reward gained by the AAV as it transitions from state s_t to s_{t+1} by taking action $a =< l, t >$, denoted as $R(s_t, < l_i, t_j >) = r_{t+1}$. Figure 2 illustrates an example of converting a AAV parcel delivery through public transportation vehicles into a markov decision model, highlighting the transitions and actions involved in the decision-making process.

C. The Idea of Q-Learning

Building on the MDP framework, we next employ Q-learning as the core reinforcement learning algorithm to address the optimization of AAV path planning. While dynamic programming techniques, such as value iteration and policy iteration, are commonly used for offline path planning by relying on pre-existing data, they become inefficient in dynamic environments where real-time adjustments are necessary. This is particularly true in scenarios where uncertainties, time-dependent public transportation schedules, fluctuating connectivity quality, and limited AAV information must be accounted for.

By leveraging the MDP model, we implement the Q-learning algorithm for multi-objective optimization of the AAV's path. Q-learning enables the AAV to learn optimal policies in an online, adaptive manner by interacting with its environment. The objectives in our case include minimizing delivery time, energy consumption, and improving connectivity quality. By adapting to changes in public transportation schedules and fluctuations in connectivity quality, Q-learning helps the AAV autonomously optimize its path in real-time, considering the constraints of limited battery capacity.

In the following section, we describe how Q-learning is applied to this problem. Initially, possible solutions are derived offline by varying the values of γ (discount factor) and δ (learning rate). Once the decision-making agent selects the best values for γ and δ, these parameters remain fixed during the online operation of the AAV. As the AAV navigates its route, it continuously adjusts its path in response to real-time environmental changes, using the pre-determined values of γ and δ.

Q-learning is particularly effective for this problem because it allows for the optimization of AAV flight paths in a dynamic and uncertain environment. Unlike traditional methods that rely on an explicit model of the environment, Q-learning learns through trial and error, accumulating valuable experience to improve the AAV's performance over time. The agent's performance is measured by a reward metric, which reflects the objectives being optimized—such as reduced travel time, energy efficiency, and improved connectivity.

Moreover, the choice of Q-learning is motivated by its computational efficiency, particularly in dynamic and real-time environments. Unlike traditional path planning algorithms, such as A*, which may require recalculating the entire path upon environmental changes, Q-learning is able to adapt in real-time without needing to recompute the full path. This ability to dynamically adjust the agent's decision-making, coupled with its relatively lower computational overhead in comparison to more complex models like Deep Q-Networks (DQN), makes Q-learning an ideal candidate for multi-objective path planning. Moreover, Q-learning's model-free nature allows it to efficiently handle varying state-action spaces and conflicting objectives, such as minimizing delivery time and energy consumption, making it a flexible and scalable solution.

In our study, the AAV is modeled as the agent navigating between a warehouse and a customer. The path planning system integrates AAV flights with public transportation, with the aim of minimizing key metrics, including travel time, energy consumption, and maintaining high connectivity quality.

V. Q-Learning for Multi-Objectives Path Planning

In this section, we provide a comprehensive explanation of how Q-learning is applied to address the multi-objective path planning problem. We focus on the specific Q-learning functions used to tackle the optimization challenge, detailing their role in minimizing the various objectives involved.

The reward function assumes a pivotal role as a real-time performance metric. Following each action taken by the AAV, such as the selection of a specific trip, the environment promptly supplies feedback pertaining to that particular choice. This real-time feedback serves as a cornerstone for the assessment of the AAV's actions, facilitating a precise evaluation of their performance. The development of the reward function is a collaborative effort, with contributions from both the environment and the decision-maker. It encompasses several critical factors, including energy consumption, connectivity quality, and considerations related to waiting and traversal times. This multifaceted function is expressed mathematically as:

$$r=\begin{cases}\gamma(W_t+\theta^{l_it_t}_{vv'})+(1-\gamma-\partial)\varepsilon^{l_it_t}_{vv'}+\partial e_h\times W_t & \text{if } l_i\neq l_0\\ \gamma\theta^{l_it_t}_{vv'}+(1-\gamma-\partial)\varepsilon^{l_it_t}_{vv'}+e_f\times\theta^{l_it_t}_{vv'} & \text{if } l_i=l_0\end{cases} \tag{6}$$

In this equation, the term W_t signifies the hovering time on node v at the current time, (here current state $S_t=(v,t)$). The computation is as follows:

$$W=\tau(t_j,v)-\alpha(t_{j-1},v) \tag{7}$$

Here, $\alpha(t_{j-1})$ denotes the instant of arrival of the previous trip t_{j-1} at node v and $\tau(t_j,v)$ represents the instant of departure of trip t_j at node v.

The formulation $l_i\neq l_0$ indicates that the AAV relies on public transportation for its action.

In this context, the reward function takes into account various factors, including waiting times, traversal times, connectivity quality, and energy consumption during hovering, while excluding considerations regarding energy consumption during flying. Conversely, when $l_i=l_0$, it indicates that the AAV is embarking on a flying travel action. In this scenario, the reward function encompasses traversal times, connectivity quality, and energy consumption exclusively during flying periods.

Similarly, the Q-value function plays a pivotal role in the Q-Learning methodology for problem-solving, particularly when modeled as a MDP. In our research, we initiate the Q-value with random values at the outset of the learning process. Consequently, as learning progresses and with each action selection, we employ the Bellman equation [35] to update the estimated Q-value:

$$\begin{aligned}Q_{new}(s_t,a)&=Q_{old}(s_t,a\rangle\\&\quad+\lambda\left[r+\sigma\min_{a'}Q_{old}\left(s_{t+1},a'\right\rangle-Q_{old}(s_t,a)\rangle\right]\end{aligned} \tag{8}$$

- Here, a the current action and a' the anticipated future action.
- $Q_{old}\left(s_{t+1},a'\right\rangle$ signifies the previous value.
- $\max_{a'}Q_{old}\left(s_{t+1},a'\right\rangle$ represents an estimate of the optimal future value.
- λ and σ are the learning rate and discount factor of the Q-learning algorithm, respectively

Upon the completion of the training process, the optimal Q-value function $Q^*(s,a_t)$, as outlined in equation 9, signifies the maximum action-value function across all policies.

$$Q^*(s_t,a_t)=\max_{a'}Q(s_{t+1},a') \tag{9}$$

To ensure a balanced exploration and exploitation strategy during the deployment of the Q-learning model, we implement the ϵ-greedy approach. In simple terms, this involves generating a random value within the range $[0,1[$, subsequently comparing it to a predefined ϵ value (e.g., $\epsilon=0.1$ in our work). If the generated value surpasses ϵ, we opt for the action corresponding to the maximum Q-value, rather than selecting an action at random. This strategy effectively mitigates the risk of getting stuck in local optima. The details of the Q-learning algorithm are summarized in Algorithm 1.

Algorithm 1 Q-Learning for AAV Path Planning

```
Initialize Q-value function Q(s, a) randomly for all states
  s and actions a
Set learning rate λ (e.g., 0.1)
Set discount factor σ (e.g., 0.9)
Set exploration rate ϵ (e.g., 0.1)
for each episode do
    Initialize state s_t from the starting state
    while episode is not finished do
        Generate a random value r ∈ [0, 1)
        if r > ϵ then
            Exploitation: Choose action a =
              arg max_a Q(s_t, a) for all actions a
        else
            Exploration: Choose a random action a
        end if
        Execute action a and observe new state and reward
          (s_{t+1}, r) = perform_action(s_k, a)
        Update Q-value using the Bellman equation 8
        Move to the next state: s_t = s_{t+1}
    end while
end for
After training, derive the optimal policy:
for each state s do
    optimal_action = arg max_a Q*(s, a) for all actions a
    Store or display optimal_action as needed
end for
```

VI. Simulation Results and Analysis

In this section, we commence by offering an overview of the simulation scenarios utilized in our experiments. Subsequently, we delve into a comprehensive performance analysis of our Q-Learning approach for path planning within a AAV-APT system optimized for multiple objectives. We conduct a comparative analysis of the behavior and performance of our proposed Q-Learning algorithm against an offline path planning method, specifically a variant of the well known A* algorithm tailored to handle our multi-objective problem. This variant of the A* algorithm excels in determining the path prior to the initiation of the delivery process. However, given the stochastic nature of the bus schedule environment, there is a risk that the A* algorithm may produce an unreliable path

TABLE II

SUMMARY OF ASSUMPTIONS FOR SIMULATION

Drones have limited battery life
Recharging of drones is not considered
Drone can only serve one parcel by trip
Cargo have the same type
Cargo has the same weight for both systems
Drones can fly directly
Drones can land/takeoff on the roof of a bus
Drones can wait for the bus at the bus station
Drones can use several buses on their trips to reach customers
Traffic conditions can change during a bus trip
Wind conditions are neglected for drones trips
drones can fly at different altitudes
Buses equipped with docking platforms compatible with AAV

to the customer. For instance, if we take a delayed bus, we may miss our journey with the next bus included in calculated path, which explains why A* solutions are unreliable.

To mitigate this risk, we incorporate an additional mechanism into A* designed to repair a broken path during the delivery process. Our evaluation encompasses a range of critical metrics, including the arrival times of the AAV at the customer's location, energy consumption (measured as the percentage of consumed energy from the total energy of the AAV battery), the total interference accumulated throughout the path calculated from our communication model, and the success rate of delivery tasks, calculated as the percentage of tasks successfully completed. A successful delivery task entails the AAV arriving at the customer's location without an empty battery and without surpassing the interference threshold that causes interruption of AAV connectivity with infrastructure. Moreover, our comparison of various algorithms is made by variations of different weight values in multi-objective function optimization, observing how the optimization algorithm behaves under different trade-off scenarios. Finally, we delve into a detailed discussion and analysis of the performance achieved by our proposed method. To accurately evaluate the proposed system, several assumptions were made to define the simulation framework. These assumptions are summarized in Table II.

A. Scenario Simulation and Environment

We commence by describing the scenarios created for our experiments. First, we reference the technical parameters of the AAV used in our simulation to the Amazon Prime Quadcopter Air delivery AAV [36]. These specifications offer a dependable foundation for establishing the key characteristics of the AAV in our analysis. You can find the detailed technical parameters of the Amazon Prime Quadcopter Air Delivery AAV in Table III. Secondly, as an example of a public transportation network, we employ a simplified network shown in Fig. 3. This network comprises five stations and five lines of public transportation. Mean and standard deviations of the link traversal and departure time instants are presented in Table IV.

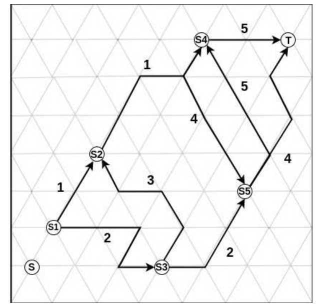


Fig. 3. Adopted public transportation network used to illustrate how the Q-learning works.

TABLE III

AMAZON PRIME QUADCOPTER AIR DELIVERY TECHNICAL PARAMETERS

Specification Parameter	Value
AAV Model	Amazon Prime Quadcopter Air Delivery
AAV Weight	3.8 kilograms
Max. Package Weight	2.3 kilograms
Max. Payload Capacity	16 kilograms
Max. Flight Time	30 minutes
Max. Speed	97 km/h
Max. Altitude	122 meters
Operating Range	16 km

TABLE IV

DISTRIBUTIONS OF LINK TRAVERSAL TIMES AND DEPARTURE INSTANTS

Line	Link	Traversal Times	Departure Instants
L1	S_1S_2	13(2), 13(2), 12(2), 11(1.5), 11(1.5)	-1(1), 9(1), 19(1), 29(1), 39(1)
	S_2S_4	10(2.5), 10(2.5), 11(2), 11(2), 10(2.5)	12(3), 22(3), 31(3), 40(2.5), 50(2.5)
L2	S_1S_3	15(1.4), 15(1.4), 16(1), 16(1)	-5(2), 2(2), 9(2), 16(1)
	S_3S_5	10(1.3), 10(1.2), 10(1), 11(1.1)	10(3.4), 17(3.4), 25(3.4), 32(2)
L3	S_3S_2	5(1), 7(1), 7(1), 6(1.5)	5(1), 17(1.5), 29(1), 41(1)
L4	S_4S_5	7(1), 8(1.2), 8(1.2), 7(1.2), 7(1)	8(3), 19(3.2), 29(2.7), 38(2.7), 48(2.5)
	S_5T	3(1), 3(1), 3(1.2), 4(1), 4(1)	8(3), 19(3.2), 29(2.7), 38(2.7), 48(2.5)
L5	S_5S_4	10(2), 10(2), 10(2), 10(2)	5(1), 20(1), 35(1.5), 50(1.5)
	S_4T	5(1), 5(1), 5(1.2), 5(0.5)	15(3), 30(3), 45(3.5), 60(3.5)

TABLE V

BASIC PARAMETERS FOR CELLULAR-BASED CONNECTED NETWORK AND AAV

Parameter	Value
Environment	Urban
Carrier frequency	2 GHz
GBS bandwidth	20 MHz
Tx power of GBS	41 dBm
Tx power of AAV	20 dBm
GBS height	25 m
Antenna Height	10 m
Channel Bandwidth (GBS)	5 MHz
Propagation Model (Height$\leq$25)	Non-line-of-sight
AAV Altitude	100 m
Path Loss Model (AAV) (Height>25)	Line-of-sight

Thirdly, from a communication perspective, we have compiled the primary simulation parameters associated with the cellular-based connected network in Table V. This table delineates vital parameters and configurations of a communication network operating within an urban environment. It furnishes essential information, including the carrier frequency (2 GHz), GBS bandwidth (20 MHz), transmission powers

TABLE VI
SIMULATION PARAMETER RANGES

Parameter	Value
Simulation Area Dimensions	3km x 2km
Number of Users (UEs)	[50 , 200]
Number of Users in Group (UEs)	[10 , 20]
Number of Group	[5 , 10]
Number of Base Stations (BSs)	[3 , 10]
Possible Positions (x, y) (km)	((0,3], (0,2])
Speed (m/s)	[0.5 , 5]
Pause Time (minutes)	[1 , 15)
Simulation Time (minutes)	1440 (1 day)
Network Handovers	Enabled

for both Ground Base Stations (41 dBm) and Autonomous Aerial Vehicles (20 dBm), GBS installation height (25 meters), and antenna height (10 meters) for both GBS and AAVs. Furthermore, it specifies the channel bandwidth for GBS communication (5 MHz) and distinguishes between non-line-of-sight and line-of-sight propagation models based on altitude, with 'Non-line-of-sight' applying to altitudes below 25 meters and 'Line-of-sight' for altitudes exceeding 25 meters. This table serves as a comprehensive reference for comprehending the fundamental communication parameters and network characteristics within the context of the cellular-based connected network and AAVs, simplifying the process of analysis and performance assessment within this specific urban environment.

Lastly, Table VI provides an overview of the parameter ranges used in our mobile network simulation, which is designed to emulate real-world scenarios. The simulation area is defined as 3 kilometers by 2 kilometers, with user equipment (UE) numbers ranging from 50 to 200, grouped in sizes of 10 to 20 UEs within 5 to 10 user groups. Simulated base stations (BSs) range from 3 to 10, with users positioned within specific coordinates. Mobility parameters include user speeds between 0.5 and 5 meters per second and pause times of 1 to 15 minutes, while the total simulation time spans a day (1440 minutes). Network handovers are enabled, creating a comprehensive model for analyzing mobile network behaviors and performance in urban environments. The simulations were conducted using Python, chosen for its robust ecosystem of libraries that facilitate numerical computation and reinforcement learning. Key libraries utilized include NumPy for mathematical operations and SimPy for event-driven simulation of the coordination between drones and public transportation. The system models and Q-learning algorithm were implemented using custom Python scripts to optimize the decision-making process for drone trajectories.

B. Performance Evaluation and Comparison

In the context of a delivery application, the variations of different weight values in multi-objective function optimization become especially relevant due to the diverse objectives involved in the delivery process. By varying the weight values assigned to each objective, we can observe how the optimization algorithm behaves under different trade-off scenarios.

To evaluate this, we compare different algorithms: Q-learning (online path planning), A* [37] (offline path planning), and A* with reparation, across various scenarios by varying the weight values in a multi-objective optimization function. These weights include parameters such as γ (importance of early arrival time), ∂ (importance of energy consumption minimization), and $(1-(\gamma+\partial))$ (importance of minimizing interference). In the first part, we vary γ to represent the importance of early arrival time, ranging from low to high values, while the remaining weights complement the sum to 1. Similarly, in the second part, we vary ∂ to represent the importance of another factor, and in the third part, we vary $(1-(\gamma+\partial))$ to reflect the importance of minimizing interference in the multi-objective optimization function. Through these variations, we aim to analyze the performance of our Q-Learning algorithm across various metrics, including timely delivery, energy efficiency, interference reduction, and success rates in computing uninterrupted paths. Additionally, comparing our algorithm with A* and A* with reparation provides insights into selecting the most suitable algorithm for specific delivery applications.

1) ***Comparison Across Multiple (γ) values:*** In a delivery application, timely delivery is often crucial. Varying γ allows us to explore how much emphasis is placed on minimizing delivery time. Lower values of γ may prioritize other factors, such as minimizing total received interference or reducing energy consumption, over speedy delivery. Conversely, higher values of γ may prioritize prompt delivery at the expense of these other considerations.

According to Table VII, increasing the value of γ from low to high results in a decrease in early arrival time for A* and A* with reparation, as well as a reduction in the range of arrival times, indicating greater closeness between them. However, this improvement in early arrival time comes at the expense of overall energy consumption, as AAVs fly more frequently and consume more energy. Additionally, the range of total received interference increases at higher γ values for these algorithms, which can be attributed to their inability to adapt to the movement of high-interference zones. Consequently, the success rate is limited to no more than 80% for A* with reparation in most scenarios and even lower for A*.

In contrast, for the Q-learning algorithm, increasing γ from low to medium-large values leads to a reduction in early arrival time, while the range of arrival times remains relatively unchanged. This stability is due to the algorithm's continued reliance on bus trips in its paths, resulting in lower energy consumption compared to A* and A* with reparation. Furthermore, the total accumulated interference is significantly lower, as the algorithm accounts for the movement of high-interference zones. However, assigning a high value to γ narrows the range of early arrival times towards the higher end of the medium-large spectrum, with energy and interference levels remaining lower than those of A* and A* with reparation.

Q-learning outperforms other algorithms in terms of energy consumption, total interference, and success rate, while offering comparable early arrival times. *However, it is important to note that the arrival times cited in the table and used in our comparison include broken paths. As a result, A* and A* with reparation may appear to outperform our algorithm*

TABLE VII

COMPARISON AND ANALYSIS OF ALGORITHM PERFORMANCE ACROSS MULTIPLE γ VALUES IN MULTI-OBJECTIVE FUNCTIONS

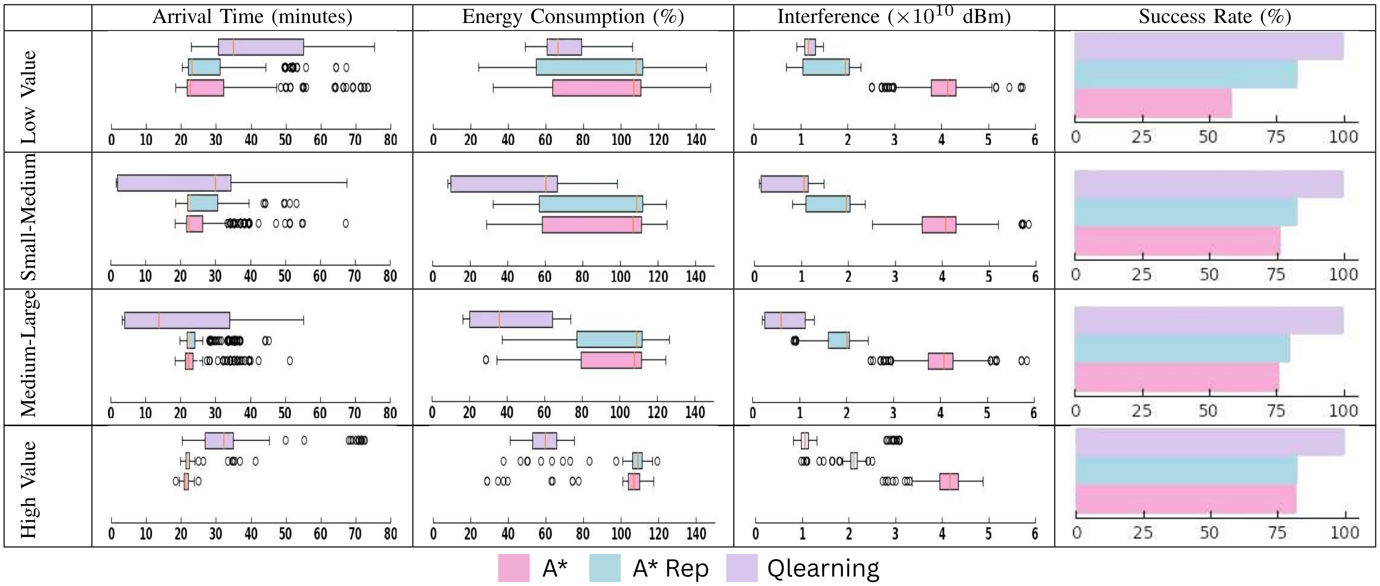
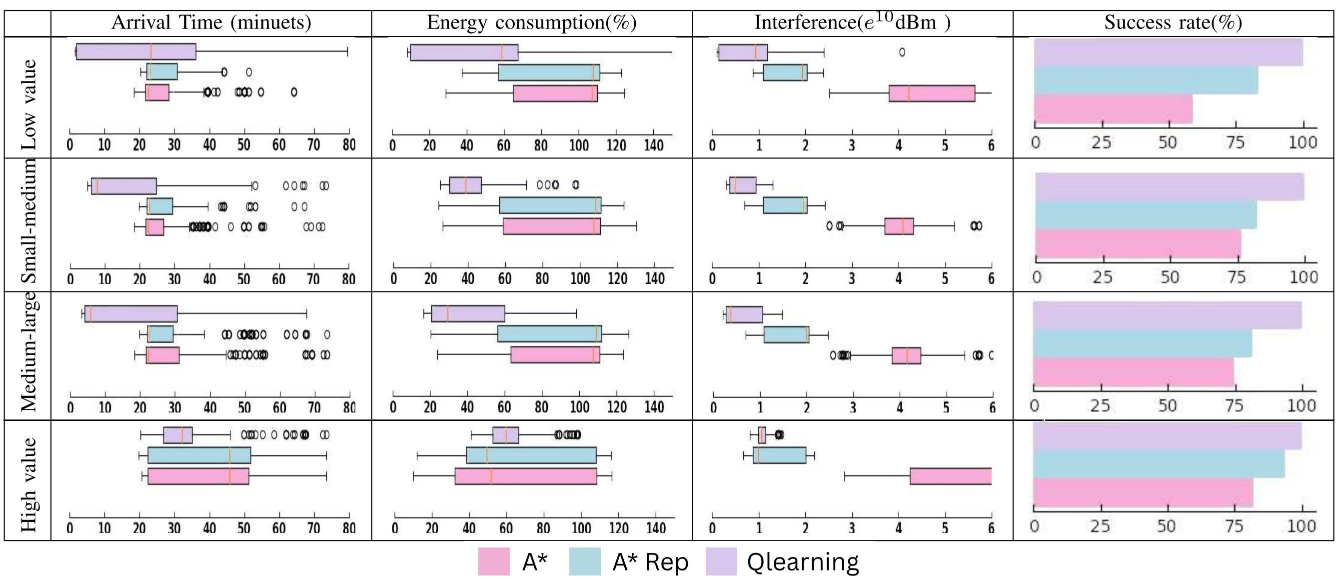


TABLE VIII

COMPARISON AND ANALYSIS OF ALGORITHMS PERFORMANCE ACROSS MULTIPLE (∂) VALUES IN MULTI-OBJECTIVE FUNCTIONS



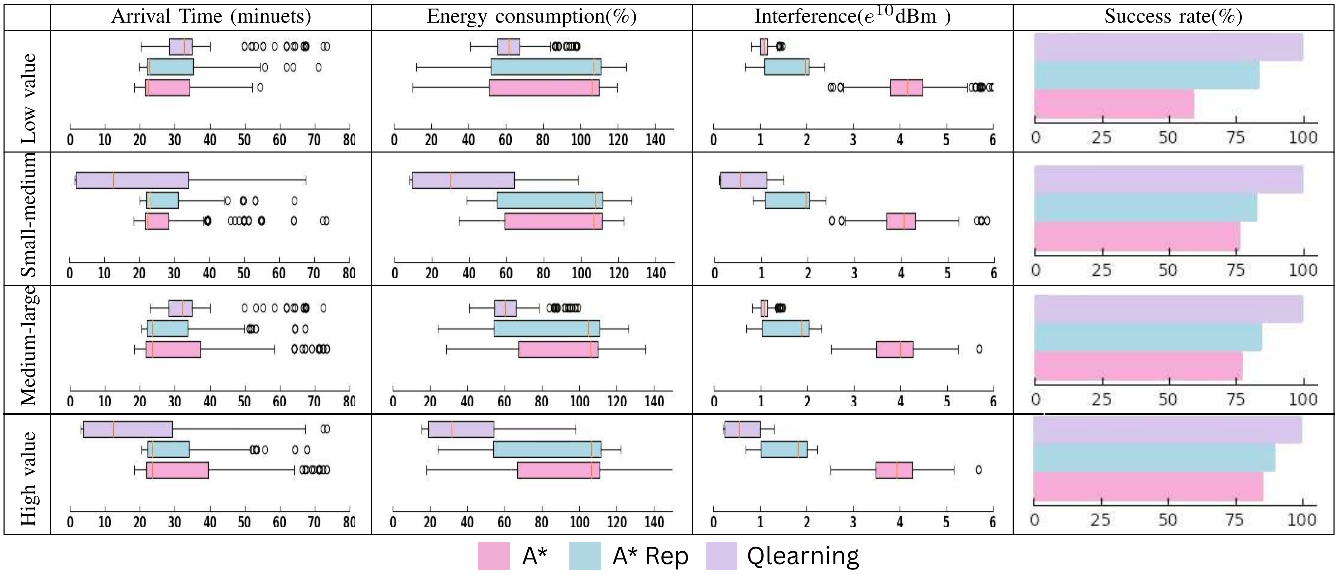
in arrival time, but this is misleading because approximately 40% and 25% of trips are unsuccessful with A and A* with reparation, respectively*. The strength of Q-learning lies in its adaptability to changes in γ within our AAV-APT delivery system. It balances the use of public transportation to minimize energy consumption and employs additional AAV flights to avoid zones with higher interference, ensuring a 100% success rate in the delivery process.

*2) **Comparison Across Multiple (∂) values**:* Energy efficiency is a critical concern, particularly in delivery operations involving vehicles. Varying the parameter ∂ allows us to analyze the trade-off between minimizing energy consumption, such as fuel usage for vehicles, and other objectives. Lower values of ∂ may prioritize energy conservation, whereas higher values might prioritize factors like delivery speed or minimizing total interference.

According to Table VIII, increasing the value of ∂ from low to medium-large values results in the same energy consumption for A* and A* with reparation. This suggests that these algorithms, lacking the experience of selecting between AAV flights and bus trips, yield identical energy consumption across these different values. Conversely, a decrease in energy consumption at higher values with spacing in the given range can be attributed to the AAV predominantly opting for bus trips to reach its destination, albeit at the expense of increased early arrival time. Furthermore, the range of total received interference increases at higher ∂ values, indicating these

TABLE IX

COMPARISON AND ANALYSIS OF ALGORITHMS PERFORMANCE ACROSS MULTIPLE $(1-(\gamma+\partial))$ VALUES IN MULTI-OBJECTIVE FUNCTIONS



algorithms' inability to adapt to areas with high interference zones.

Moreover, we observe an increase in the success rate of the delivery process from 55%-75% to 80%-90% for A* and A* with reparation due to the closer correspondence between power consumption and successful delivery. The remaining gaps in success rates can be attributed to the total interference received by the AAV exceeding the threshold required for the AAV to remain connected.

Conversely, for the Q-learning algorithm, an increase in ∂ from low to medium-large values leads to a decrease in energy consumption, altering the range of values. This reflects Q-learning's ability to leverage its experience in selecting between AAV flights and bus trips, resulting in different energy consumption for these values. Although there is no significant decrease at large values, a tendency towards medium-large values suggests that the AAV adds another AAV flight to avoid high interference zones, ensuring continued connectivity.

Q-learning outperforms other algorithms across all metrics by varying ∂. Its superiority lies in its adaptability to changes in ∂ within our AAV-APT delivery system, balancing the use of public transportation to minimize energy consumption and employing more AAV flights to evade areas with higher interference levels, ultimately ensuring a 100% success rate in the delivery process.

*3) **Comparison Across Multiple** $((1-(\gamma+\partial)))$ **values**:* Minimizing interference from other users and base stations is critical for maintaining reliable communication links and safe AAV operations. Varying $(1-(\gamma+\partial))$ directly assesses the importance of interference mitigation relative to other objectives. A higher weight on interference minimization indicates a greater emphasis on ensuring minimal disruption from other sources, even if it comes at the expense of early arrival time or energy consumption.

According to Table IX, it's evident that increasing the value of $(1-(\gamma+\partial))$ from low to high results in the same interference level for both A* and A* with reparation. Additionally, when γ and ∂ go from 0.5 to 0 in different scenarios, the range of total consumed energy and early arrival time remains consistent. This suggests that these algorithms lack the experience and adaptability to avoid areas with high interference zones, resulting in identical interference levels across different values. Moreover, there's an observable increase in the success rate of the delivery process from 55%-75% to 80%-90% for A* and A* with reparation, respectively. This increase is attributed to the higher importance placed on minimizing interference, allowing AAVs to remain connected and complete more delivery processes. However, the remaining gaps in success rates may be attributed to AAVs exceeding the capacity of battery energy (100%) while attempting to avoid areas with high interference. Conversely, for the Q-learning algorithm, an increase in $(1-(\gamma+\partial))$ from low to high values leads to a decrease in interference levels, altering the given value ranges. This indicates that Q-learning leverages its experience to effectively avoid zones with high interference while conserving much more AAV energy and maintaining comparable early arrival times.

Q-learning outperforms other algorithms across all metrics by varying $(1-(\gamma+\partial))$. Its superiority lies in its adaptability to changes within our AAV-APT delivery system, effectively avoiding zones with high interference levels by selecting trips with low interference and adding more AAV flights to evade areas with higher interference levels, thereby ensuring a 100% success rate in the delivery process.

We fully acknowledge the importance of real-world testing to validate the robustness and reliability of the proposed method in dynamic and unpredictable urban environments. As part of our future work, we intend to extend the evaluation of our approach through real-world experiments in collaboration with urban logistics companies and transportation authorities. These experiments will enable us to assess the system under practical conditions, including fluctuating traffic patterns, variable weather conditions, and operational constraints, and to refine the method based on empirical observations. Nevertheless, the simulation environment employed in this study incorporates significant real-world complexities, such as stochastic delays in public transportation schedules, fluctuating energy levels, and interference scenarios. The demonstrated consistency and robustness of our Q-network-based approach in addressing these uncertainties within the simulation strongly suggest its potential effectiveness in real-world applications.

C. Discussion and Analysis

Based on the results from the preceding sections, significant differences were observed in algorithm performance across various metrics and weight values. Here's a concise analysis:

- Q-learning consistently achieved competitive early arrival times compared to A* and A* with reparation, suggesting its effectiveness in balancing prompt delivery with energy consumption and interference minimization.
- Q-learning achieved a 100% success rate in delivering packages across all tested scenarios, demonstrating its robustness and reliability in handling dynamic conditions and ensuring timely deliveries, even under unpredictable changes in public transport schedules and user positions.
- Q-learning demonstrated superior energy efficiency, particularly at higher weight values, by effectively leveraging AAV flights and bus trips to minimize energy consumption.
- Q-learning outperformed other algorithms in minimizing interference levels, ensuring reliable communication links and safe AAV operations.
- The adaptability of Q-learning led to consistently superior performance across all metrics and weight values compared to A* and A* with reparation.

In conclusion, Q-learning emerges as the most suitable algorithm for optimizing delivery operations in AAV-APT systems due to its ability to balance efficiency, reliability, and connectivity safety effectively. While A* and A* with reparation showed competitive performance in specific scenarios, they couldn't match Q-learning across all evaluated metrics.

VII. Conclusion

In this paper, we investigated the effectiveness of Q-learning for online multi-objective path planning within AAV-APT systems. This research contributes to the general body of knowledge by addressing a clear research gap: the application of adaptive reinforcement learning techniques for dynamic, real-time multi-objective optimization in AAV-APT delivery, which remains underexplored in the literature. Through comprehensive simulations and performance analyses, we demonstrated that Q-learning significantly outperforms traditional offline methods such as variants of the A* algorithm, achieving superior efficiency, reliability, and adaptability. Our findings emphasize Q-learning's ability to balance multiple objectives—energy consumption, interference avoidance, and timely delivery—resulting in a 100% package delivery success rate and consistent superiority across all evaluated metrics. These key outcomes offer important insights that can guide stakeholders, including urban logistics planners, AAV operators, and the research community, in adopting reinforcement learning-based approaches to improve last-mile delivery operations in urban environments.

While the proposed Q-learning approach shows clear originality and strong practical significance, it has some limitations. Specifically, it requires substantial training data and computational resources, and the manual tuning of multi-objective weighting parameters may limit adaptability to changing priorities. To mitigate these limitations, future research should focus on hybrid methods that combine the strengths of different algorithms to enhance performance and scalability. Additionally, developing adaptive mechanisms to automatically adjust weighted values tailored to delivery application specifics will improve flexibility. Furthermore, extending this work to multi-AAV path planning represents an important direction for scaling delivery systems and coordinating multiple drones effectively. Overall, this study strongly contributes to advancing the state of the art by demonstrating Q-learning as a promising and practical solution for multi-objective path planning in AAV-APT systems, balancing efficiency, reliability, and safety.

Appendix

A. Communication Model

1) Thermal Noise: To calculate the value of the thermal noise spectral power δ^2, various factors are considered, including the bandwidth of the communication channel and the temperature of the environment. Therefore, the formula for thermal noise power density is:

$$\delta^2 = k \cdot T \cdot B \tag{10}$$

where δ^2 is based on the Boltzmann constant k, approximately equal to 1.38×10^{-23}, T is the absolute temperature in (K), and B is the bandwidth of the communication channel in (Hz).

2) Large-Scale Slow Fading: In wireless communication, various path loss models and large-scale fading models are employed to calculate the value of large-scale slow fading, represented as (L_m^t), which depends on the specific characteristics of the environment and propagation conditions. In our study, where the focus is on developing a delivery system in urban areas, we utilize the ITU urban and suburban path loss model [38] to estimate large-scale path loss in urban and suburban environments. Therefore, path loss models typically follow a formula like:

$$L(d) = L_0 + 10n \log_{10}(d) + X_f + \chi \tag{11}$$

where L_0 represents the path loss at a reference distance, typically set to 1 meter and at a reference frequency. The parameter n serves as the path loss exponent, determining the rate at which path loss increases with distance. The term X_f accommodates frequency-dependent effects on path loss. Lastly, χ is a log-normal random variable introduced to account for shadow fading and environmental variability. (see [38] which offer comprehensive guidance on values of these model parameters)

3) Small-Scale Fast Fading: As our study focuses on developing a delivery system in urban areas, radio wave propagation is significantly influenced by factors such as the presence of buildings, moving vehicles, and various obstacles. Consequently, we employ the Rayleigh fading model to characterize our F_m^t component. This model is particularly well-suited for capturing the effects of non-line-of-sight (NLOS) scenarios commonly encountered in urban environments. The probability density function (PDF) for the magnitude of $F_m(t)$ can be expressed as:

$$f(x) = \frac{\sigma^2}{2x} e^{-\frac{\sigma^2 x^2}{2}} \tag{12}$$

where x is the magnitude and σ is the standard deviation in the urban environment.

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Mohammed Rahmani received the master's degree in distributed networks and systems from Ferhat Abbas University, Algeria, in 2019, and the Ph.D. degree in computer science from the University of Burgundy, France, in 2024. He is a Post-Doctoral Researcher at the CNRS GREYC Laboratory, University of Caen, France. His research interests include the Internet of Things (IoT), intelligent transportation systems, systems engineering, AI-based cybersecurity, and various applications in computer science.

Sidi Mohammed Senouci (Senior Member, IEEE) received the Ph.D. degree in computer science from the University of Paris 6 in 2003 and the H.D.R. degree from INP Toulouse, France, in 2009. From 2004 to 2010, he was a Researcher at Orange Labs, Lannion. Since September 2010, he has been a Professor at the University of Bourgogne. Since October 2017, he has been the Director of the DRIVE Laboratory, Nevers. He participated in or still participates in several national and European-wide research projects (ITEA CarCoDe, ITEA FUSE-IT, ITEA PARFAIT, Visio3D+, ANR 5G-INSIGHT, and OPEVA), holds seven international patents, and has published his work in major IEEE conferences and renowned journals. He was the Co-Chair of different IFIP, ACM, and IEEE conferences and workshops. He was the Chair of IEEE ComSoc IIN Technical Committee, TCIIN (2014–2016). He is part of the Editorial Board of *IEEE Network Magazine* and serves as a Guest Editor of premium journals. He is also a Senior Expert in the societies IEEE and the Society of Electricity and Electronics (SEE).

Florian Delavernhe received the Ph.D. degree from the University of Angers, France, in 2020, under the supervision of Prof. André Rossi and Prof. Marc Sevaux. He had a short stay at Massachusetts Institute of Technology (MIT), where he worked with Prof. Patrick Jaillet. Afterwards, he worked for one year post-doctoral researcher on a joint project between the University of Angers and HEC Montréal, with Prof. Jorge E. Mendoza. After the completion of his post-doctoral research, he became an Associate Professor at the University of Burgundy and a member of the DRIVE Laboratory, where he joined the Systèmes Intelligents et Connectés (SIC) Component. His research focuses on operations research and particularly on the modeling and solving of complex optimization problems. More precisely, his research curiosity is sparked by connected systems (e.g., drones or sensor networks) and the target search problem.

Marion Berbineau (Member, IEEE) received the Engineering degree in informatics, electronics, and automatics from Polytech Lille formerly EUDIL in 1986 and the Ph.D. degree in electronics from the University of Lille, France, in 1989. She has been the Research Director of the Université Gustave Eiffel (previously IFSTTAR) since 2000. She was the Director of the LEOST Laboratory from 2000 to 2013, and then the Deputy Director of the COSYS Department from 2013 to 2017. In addition to research activities and supervision of Ph.D. Students, she coordinates railway research at Université Gustave Eiffel. She is the Pole Leader of the Intelligent Mobility Pole of European Railway Research Network of Excellence (Eurnex). Her current interests include wireless communications for connected and automatic vehicles (trains and cars) (radio propagation, channel characterization and modelling, SDR, MIMO, ITS-G5, GSM-R, LTE, 5G NR, and beyond). She has already participated in a lot of European and national research projects. She is an author and co-author of several journal articles and patents.